Lab4 polynomial regression

Jupyter Notebook — original cells shown separately and in order

CELL 01 | CODE | EXECUTION COUNT: 20

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder,StandardScaler
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import accuracy_score, classification_report
from sklearn.metrics import mean_squared_error, r2_score

import warnings
warnings.filterwarnings('ignore')
```

CELL 02 | CODE | EXECUTION COUNT: 21

```
df=pd.read_csv("insurance.csv")
```

CELL 03 | CODE | EXECUTION COUNT: 22

```
df
```

OUTPUT

```
      age     sex     bmi  children smoker     region      charges
0      19  female  27.900         0    yes  southwest  16884.92400
1      18    male  33.770         1     no  southeast   1725.55230
2      28    male  33.000         3     no  southeast   4449.46200
3      33    male  22.705         0     no  northwest  21984.47061
4      32    male  28.880         0     no  northwest   3866.85520
...   ...     ...     ...       ...    ...        ...         ...
1333   50    male  30.970         3     no  northwest  10600.54830
1334   18  female  31.920         0     no  northeast   2205.98080
1335   18  female  36.850         0     no  southeast   1629.83350
1336   21  female  25.800         0     no  southwest   2007.94500
1337   61  female  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]
```

CELL 04 | CODE | EXECUTION COUNT: 23

```
df.info()
```

OUTPUT

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

CELL 05 | CODE | EXECUTION COUNT: 24

```
df.describe
```

OUTPUT

```
<bound method NDFrame.describe of       age     sex     bmi  children smoker     region      charges
0      19  female  27.900         0    yes  southwest  16884.92400
1      18    male  33.770         1     no  southeast   1725.55230
2      28    male  33.000         3     no  southeast   4449.46200
3      33    male  22.705         0     no  northwest  21984.47061
4      32    male  28.880         0     no  northwest   3866.85520
...   ...     ...     ...       ...    ...        ...         ...
1333   50    male  30.970         3     no  northwest  10600.54830
1334   18  female  31.920         0     no  northeast   2205.98080
1335   18  female  36.850         0     no  southeast   1629.83350
1336   21  female  25.800         0     no  southwest   2007.94500
1337   61  female  29.070         0    yes  northwest  29141.36030

[1338 rows x 7 columns]>
```

CELL 06 | CODE | EXECUTION COUNT: 25

```
df.shape
```

OUTPUT

```
(1338, 7)
```

CELL 07 | CODE | EXECUTION COUNT: 26

```
df.dtypes
```

OUTPUT

```
age           int64
sex          object
bmi         float64
children      int64
smoker       object
region       object
charges     float64
dtype: object
```

CELL 08 | CODE | EXECUTION COUNT: 27

```
df.head()
```

OUTPUT

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

CELL 09 | CODE | EXECUTION COUNT: 28

```
df.tail()
```

OUTPUT

	age	sex	bmi	children	smoker	region	charges
1333	50	male	30.97	3	no	northwest	10600.5483
1334	18	female	31.92	0	no	northeast	2205.9808
1335	18	female	36.85	0	no	southeast	1629.8335
1336	21	female	25.80	0	no	southwest	2007.9450
1337	61	female	29.07	0	yes	northwest	29141.3603

CELL 10 | CODE | EXECUTION COUNT: 29

```
df.notnull()
```

OUTPUT

	age	sex	bmi	children	smoker	region	charges
0	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True
4	True	True	True	True	True	True	True
...	...	...	...	...	...	...	...
1333	True	True	True	True	True	True	True
1334	True	True	True	True	True	True	True
1335	True	True	True	True	True	True	True
1336	True	True	True	True	True	True	True
1337	True	True	True	True	True	True	True

[1338 rows x 7 columns]

CELL 11 | CODE | EXECUTION COUNT: 30

```
df.isnull().sum()
```

OUTPUT

```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

CELL 12 | CODE | EXECUTION COUNT: 31

```
from scipy import stats

rm_z_scores = np.abs(stats.zscore(df['charges']))
outliers_z = df[rm_z_scores > 3]
print(f"Z-Score Method found {len(outliers_z)} outliers in the 'charges' column.")
```

OUTPUT

```
Z-Score Method found 7 outliers in the 'charges' column.
```

CELL 13 | CODE | EXECUTION COUNT: 32

```
label=LabelEncoder()
df['sex']=label.fit_transform(df['sex'])
df['smoker']=label.fit_transform(df['smoker'])
df['region']=label.fit_transform(df['region'])
df
```

OUTPUT

```
      age  sex     bmi  children  smoker  region      charges
0      19    0  27.900         0       1       3  16884.92400
1      18    1  33.770         1       0       2   1725.55230
2      28    1  33.000         3       0       2   4449.46200
3      33    1  22.705         0       0       1  21984.47061
4      32    1  28.880         0       0       1   3866.85520
...   ...  ...     ...       ...     ...     ...          ...
1333   50    1  30.970         3       0       1  10600.54830
1334   18    0  31.920         0       0       0   2205.98080
1335   18    0  36.850         0       0       2   1629.83350
1336   21    0  25.800         0       0       3   2007.94500
1337   61    0  29.070         0       1       1  29141.36030

[1338 rows x 7 columns]
```

CELL 14 | CODE | EXECUTION COUNT: 33

```
X = df[['bmi']]
y = df['charges']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training Samples :", len(X_train))
print("Testing Samples :", len(X_test))
```

OUTPUT

```
Training Samples : 1070
Testing Samples : 268
```

CELL 15 | CODE | EXECUTION COUNT: 34

```
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_state=42)
```

CELL 16 | CODE | EXECUTION COUNT: 35

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

degrees = [1, 2, 3, 4, 5, 6]
results = []  # Keep this initialization here

plt.figure(figsize=(18, 10))
x_grid = np.linspace(X.min(), X.max(), 300).reshape(-1, 1)

for degree in degrees:
    poly = PolynomialFeatures(degree)
    X_poly_train = poly.fit_transform(X_train)

    model = LinearRegression()
    model.fit(X_poly_train, y_train)

    y_pred = model.predict(poly.transform(X_test))

    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)

    # REMOVED 'results = []' from here to avoid overwriting
    results.append([degree, mse, rmse, r2])

    plt.subplot(2, 3, degree)
    plt.scatter(X, y, color='red', s=10)
    plt.plot(x_grid, model.predict(poly.transform(x_grid)), color='blue', linewidth=2)
    plt.title(f"Degree {degree}")

# Moved outside the for loop to display all 6 subplots at once
plt.tight_layout()
plt.show()
```

OUTPUT

```
<Figure size 1800x1000 with 6 Axes>
```

CELL 17 | CODE | EXECUTION COUNT: 36

```
results_df=pd.DataFrame(results,columns=["Degree","MSE","RMSE","r2 Score"])
results_df
```

OUTPUT

	Degree	MSE	RMSE	r2 Score
0	1	1.490851e+08	12210.039191	0.039702
1	2	1.504728e+08	12266.733504	0.030763
2	3	1.524525e+08	12347.166300	0.018011
3	4	1.531528e+08	12375.489932	0.013501
4	5	1.526792e+08	12356.343313	0.016551
5	6	1.538510e+08	12403.667082	0.009003

CELL 18 | CODE | EXECUTION COUNT: 37

```
plt.figure(figsize=(8,5))
plt.plot(results_df["Degree"],results_df["r2 Score"],marker='o')
plt.xlabel("Polynomial Degree")
plt.ylabel("R² Score")
plt.title("Degree vs R² Score")
plt.grid(True)
plt.show()
```

OUTPUT

```
<Figure size 800x500 with 1 Axes>
```

CELL 19 | CODE | EXECUTION COUNT: 38

```
plt.figure(figsize=(8,5))
plt.plot(results_df["Degree"],results_df["RMSE"],marker='o')
plt.xlabel("Polynomial Degree")
plt.ylabel("RMSE")
plt.title("Degree vs RMSE")
plt.grid(True)
plt.show()
```

OUTPUT

```
<Figure size 800x500 with 1 Axes>
```

CELL 20 | CODE | EXECUTION COUNT: 38

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